

No Free Credit: How Much Simulated Evidence Can Replace Real Events in Zero-Failure Reliability Demonstration of Perception Systems

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Abstract

A perception system on a new fixed camera is often accepted by a zero-failure reliability demonstration: a fixed number of real events, the zero-failure sample size (59 events for 95% confidence that at most 5% of events are missed), must pass without a miss. How many can simulated evidence replace? We prove that, without an assumption linking the simulator to reality, independent simulated data cannot shorten the demonstration; that accepting a larger error probability buys an exactly computable number of events; and that a synthetic twin paired with each real event bounds the real miss rate by the twin miss rate plus a discordance rate, which pays off only through a transfer bound priced in calibration scenes. On 48 fixed-camera scenes, real misses are rare at the pre-specified operating point (0.9%), so it is untested. If calibration carries over, a twin run about 28 times per camera would save at most 14 events each at a 20% target; the twin as built saves none. The transfer certificate stands at scene level but is withdrawn at site level (17 sites). Two named failure modes: night and camera-jitter cameras pass through (falsely accepted: 4 of 6; 1 of 26, a real-shake scene).

Keywords: reliability demonstration, zero-failure testing, synthetic data, digital twin, success-run, perception system, transfer bound

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1. Introduction

A reliability demonstration test (RDT) accepts a product when it passes a planned test; for attribute (pass/fail) data the classical plan is the zero-failure, or success-run, test [1, 2]. If each demand fails independently with probability p , a unit with $p > \varepsilon$ passes n demands with probability at most $(1-\varepsilon)^n$, so passing $n_0(\varepsilon, \delta) = \lceil \ln \delta / \ln(1-\varepsilon) \rceil$ demands without failure demonstrates $p \leq \varepsilon$ at confidence $1 - \delta$: 59 demands at $\varepsilon = \delta = 5\%$. Zero-failure planning remains common industrial practice [3], and its sample-size burden is a recurring motive for test-plan research in this journal, from success-failure verification tests [4] to qualification plans that borrow auxiliary information [5, 6].

We consider a learned perception component, a fixed object detector on a fixed camera that samples frames periodically, as the unit under test. A demand is the passage of an object (an *event*); a failure is an event on which no sampled frame yields a detection. Such components are increasingly assessed with reliability-engineering tools [7, 8, 9]. For them, the demands are rare and every new installation is a new unit: 59 failure-free events must be collected per camera, and for road vehicles the exposure needed to demonstrate rare-failure reliability by field operation alone is prohibitive [10].

Simulation is the obvious remedy. Digital twins and simulators already supply reliability evidence in engineering practice [11, 12, 13], and virtual scenario testing is a leading proposed alternative to mileage-based demonstration of automated vehicles [14, 15]. For perception, however, the simulation-to-reality gap is measurable and has been measured under different environmental conditions [16, 17], and performance in simulation need not predict performance in the field [18]. The question for a demonstration test is therefore quantitative: *how many of the n_0 real demands can simulated evidence replace, and what form of guarantee survives when the evidence is transferred to an untested unit?*

This paper answers the question for zero-failure demonstration of a perception component.

1. *No free credit.* Without an assumption that links the simulator to reality, no valid procedure can use independent simulated data to shorten a zero-failure demonstration, for any simulator and any amount of simulation. This makes precise, for success-run tests, a principle known from clinical borrowing [19], and relates to the limits on combining evidence in software dependability [20].

- 38 2. *The exact price of relaxing confidence.* Accepting error probability $\delta' >$
39 δ when the simulator is wrong buys at most, and exactly, $n_0(\varepsilon, \delta) -$
40 $n_0(\varepsilon, \delta')$ demands: 8, 14, 22 and 27 of the 59 for $\delta' = 7.5, 10, 15$ and
41 20%.
- 42 3. *A paired-twin bound and a fleet-level transfer bound.* A synthetic twin
43 of each real event bounds the real failure probability by the twin failure
44 probability plus a discordance rate, with no assumption; on one camera
45 this costs more real events than the plain test (72–122). Calibrated
46 once across scenes, the discordance transfers to an untested camera as
47 a population-average bound whose price is the number of calibration
48 scenes, not the number of events.
- 49 4. *An exchange rate on public benchmarks, and a named failure mode.*
50 On 48 fixed-camera scenes the twin as built saves no real event; a twin
51 run to convergence would save at most 14 real events per camera, if
52 the discordance calibrated with paired twins also held for the twins of
53 new cameras (not verified), at the operating point fixed in advance,
54 where real failures are rare (0.88%) and the demonstration is therefore
55 untested. The transfer certificate stands at scene level but is withdrawn
56 when the unit is the site (17 sites). When out-of-domain cameras are
57 judged alone, they pass through: 4 of 6 night cameras and, at a fixed
58 sampling phase, 1 of 26 camera-jitter cameras (a scene with real shake)
59 are falsely accepted; we call this *out-of-domain pass-through*, with two
60 named modes, night and camera jitter.

61 *Reliability demonstration tests in the zero-failure regime have no estab-*
62 *lished rule for how much simulated evidence may substitute for real evidence,*
63 *nor a guarantee form that survives transfer to an untested unit; this paper*
64 *gives a finite-sample answer (none without an assumption; an exact price*
65 *for relaxing confidence; a fleet-level bound whose price is scene count) and*
66 *measures it on 48 fixed-camera scenes.*

67 2. Related work and research gap

68 Table 1 positions the paper against four bodies of work.

69 (a) *Reliability demonstration and zero-failure test planning.* The zero-
70 numerator bound and the rule of three are classical [1, 2], and the Bayesian
71 zero-failure (BAZE) test sets the required test time through a prior on the

Table 1: Research gap. Unit: what the guarantee covers. ZF: is the zero-failure (success-run) regime treated? Price: is the real-evidence value of simulation quantified? Transfer: does the guarantee reach an untested unit? Link: assumption needed between simulation and reality. Evidence: empirical basis.

Body of work	Unit	ZF	Price of simulation	Transfer	Sim-real link	Evidence
(a) Reliability demonstration, zero-failure planning [1, 21, 22, 23, 4, 5, 24]	product / unit	yes	through a prior or auxiliary data; not as real events saved	to the tested product; prior-system evidence handled conservatively	prior or similarity assumption (Bayesian); none (frequentist, no simulation)	case studies
(b) Simulation-assisted testing, digital-twin reliability [11, 12, 13, 25, 26]	system	no	fidelity weighting in fusion	not addressed (extrapolation named as open)	fidelity or calibration model	engineering case studies
(c) Simulation-augmented evaluation of perception [16, 17, 27, 15, 28, 29]	mean metric	no	gap measured; variance reduction for means	no	paired labels (rectifier) or none	perception benchmarks
(d) Finite-sample guarantees and assurance for learned perception [30, 31, 32, 33, 8]	per deployment	no	not addressed (real calibration data)	exchangeability of calibration and deployment	exchangeability	detection data sets
(e) This paper	fleet (population average); per-unit as a data requirement	yes	exact: none without a link; $n_0(\delta) - n_0(\delta')$ when relaxing; 14 events per camera at most	yes, priced by scene count ($U(J, m) = 6.05\%$, $J = 0$, $m = 48$)	none for the twin bound; scenes exchangeable (i.i.d.) for transfer, as in (d) at scene level	48 fixed-camera scenes, 114 events; failure mode measured

72 failure rate [21]. Recent work designs sample sizes by assurance [22], stud-
73 ies the limits of Bayesian zero-failure plans [3], and publishes in this jour-
74 nal demonstration and acceptance plans for degrading products [23, 34],
75 success-failure verification tests [4] and qualification plans that use subsys-
76 tem, Monte-Carlo or expert information [5, 6]. For software and autonomous
77 systems, conservative Bayesian inference uses prior knowledge, including ev-
78 idence from previous similar systems, without optimistic bias [24, 35, 36],
79 and reusing the reliability of an older system as a prior is optimistic rather
80 than conservative [37]. Bayesian borrowing quantifies how much a prior is
81 worth through its effective sample size [38] and discounts historical data

through power or commensurate priors [39, 40], and conservative frequentist bounds have been derived when the test profile differs from the operational profile [41]. *What is missing:* a frequentist statement of how much the zero-failure sample size can be reduced by simulated rather than prior-system evidence, and at what price.

(b) *Simulation-assisted testing and digital-twin reliability.* Digital twins feed reliability models with simulated operating data [11], are validated against physical experiments [12], and are discussed as maintenance tools whose suitability is still open [42]. Multi-fidelity fusion combines scarce physical tests with abundant simulation for mission-reliability prediction, weighting the simulation by its fidelity gap [13], and simulators generate test scenarios for autonomous systems [25, 14]. Model verification, validation and uncertainty quantification identify extrapolation to untested conditions as a major challenge [26]. *What is missing:* a rule that states which assumption about the simulator is needed before its output may reduce a zero-failure test, and a measured sim–real discordance that makes the assumption checkable.

(c) *Simulation-augmented evaluation of perception.* The simulation-to-reality gap of camera object detectors has been measured across day, night, fog and rain [16] and for drone detection [17]; synthetic images are used to test driving perception, with a distribution mismatch that biases test results [27], and perception digital twins reproduce field data only approximately [43]. Statistical model checking evaluates perception in simulation only [44], simulation performance can fail to predict field performance [18], and a statistical foundation for aligning synthetic and real testing outcomes has been called for [15]. Closest to our question, Sim2Val uses simulator outputs as control variates to reduce the number of real samples needed for a confidence bound on mean performance (CoRL 2025, arXiv preprint) [45], SCAPE combines limited real rollouts with simulation proxies for policy evaluation (preprint) [46], and multi-fidelity fusion weights simulation by its fidelity gap in reliability prediction [13]; none treats the zero-failure regime. Prediction-powered inference and its stratified variant combine many proxy outcomes with few real labels through a paired rectifier [28, 29]. *What is missing:* these methods estimate a mean performance or reduce its variance; none addresses the zero-failure regime, where the quantity to bound is a rare failure probability and the rectifier itself would cost almost n_0 real events per unit.

(d) *Finite-sample guarantees and assurance for learned perception.* Conformal prediction gives distribution-free finite-sample guarantees [47], cali-

brated risk control extends them to general losses [30], and they have been applied to object detection [31], keypoint-based pose estimation [48], perception for safe navigation [32] and, in this journal, anomaly detection in cyber-physical systems [33]. Reliability of machine-learning components is also modelled directly, through multi-version architectures [7], safety-assurance cases for object detectors [8] and failure-free-operation definitions under an operational profile [9]. *What is missing:* these guarantees are calibrated on real data from the deployment distribution; they do not say how much of that calibration data may be synthetic, nor how a guarantee transfers to a unit whose own data are absent.

3. Demonstration setting

Unit, demand and failure. The unit under test is a fixed camera s with a fixed detector. The camera observes *events*: maximal runs of frames that contain the target object. It samples frames with period K and an unknown phase φ , and runs the detector on the sampled frames. An event is *missed* ($M = 1$) if no sampled frame is a hit; with H the hit frames of an event and φ uniform, $\mathbb{P}(M = 1) = 1 - |H \bmod K|/K$. The miss probability of camera s is p_s , and the demonstration goal is the assurance claim $p_s \leq \varepsilon$ at confidence $1 - \delta$. $U(x, n)$ denotes the one-sided $(1 - \delta)$ Clopper–Pearson upper bound for x failures in n demands. We call an accepted assurance claim of this form a *certificate*. In the language of acceptance sampling, δ is the consumer’s risk; the producer’s risk is reported as the power of a plan at the observed miss rate. All claims concern the miss probability averaged over the uniform sampling phase; a camera deployed with one fixed, unknown phase is covered only on average over phases.

Simulated evidence and twins. A simulator produces data $Y \sim Q$. A *paired twin* renders a synthetic counterpart of each real event and runs the same detector at the same sampling phase, giving M' . We write $q = \mathbb{P}(M' = 1)$ and $D = \mathbb{P}(M = 1, M' = 0)$ (real miss, twin catch), the *discordance*. Every rate in this paper is per event; with three sprite variants per event, the twin catches an event if at least two variants do, and an event counts as discordant if it is discordant at one or more sampling phases (worst phase; an integer count, conservative with respect to the phase-averaged rate, which we give in parentheses). Table 2 collects the notation.

Table 2: Notation.

symbol	meaning
ε, δ	target miss probability; allowed probability of a false acceptance
$n_0(\varepsilon, \delta)$	zero-failure sample size $\lceil \ln \delta / \ln(1 - \varepsilon) \rceil$
$K, \varphi, d_{\min}$	sampling period, sampling phase, minimum event length (frames)
M, M'	real miss, twin miss of an event at the same phase
p_s, q_s, D_s	real miss probability, twin miss probability, discordance $\mathbb{P}(M = 1, M' = 0)$ of camera s
m, n_s, N	calibration scenes, paired events of scene s , $N = \sum_s n_s$
X, J	discordant events; scenes with at least one discordant event
$U(x, n)$	one-sided $(1 - \delta)$ Clopper–Pearson upper bound
fleet bound, population bound	$U(X, N)$ and $U(J, m)$ (Result 4)
OP*, OP-A, OP-C	operating points $(\varepsilon, d_{\min}, K) = (20\%, 32, 16)$, $(20\%, 32, 8)$, $(10\%, 32, 1)$

154 *Random model.* Scenes are drawn from a population; given a scene, events
155 are drawn i.i.d. from the scene’s event law and each receives an independent
156 uniform phase; given the event and the phase, the real and twin outcomes
157 are determined (fixed detector, fixed sprite seeds). Every per-event indicator
158 below—real miss, twin miss, discordance at a random phase, and the worst-
159 phase indicator—is therefore a Bernoulli variable, independent across events
160 given the scene. In the audit, the “truth” of a scene is its empirical event
161 law, resampled with replacement.

162 4. What simulated evidence can buy: four results

163 The four results below are stated in words; Appendix A gives the formal
164 statements, the assumptions and the proofs, each with a numerical check.

165 *Result 1 (no free credit).* Suppose the simulated data are independent of the
166 real demands and no assumption links the simulator to reality, so that for
167 every simulator law Q both a perfect unit ($p = 0$) and a unit just worse
168 than the target ($p \downarrow \varepsilon$) are possible. Then any acceptance procedure that
169 is valid at confidence $1 - \delta$ accepts a perfect unit with probability at most

Table 3: Result 2: exact credit c versus the continuous form c^* ($\varepsilon = \delta = 5\%$, $n_0 = 59$), and the two-tier plan run with the twin event pool as an independent simulator (worst case $p = \varepsilon$; closed form given the pass probability, and 100,000 Monte-Carlo runs).

δ'	7.5%	10%	15%	20%
exact credit c (demands)	8	14	22	27
saving w.r.t. n_0	14%	24%	37%	46%
continuous form c^*	7	13	21	27
false acceptance, twin pool (closed form)	6.7%	8.6%	12.3%	15.5%
Monte Carlo	6.7%	8.4%	12.2%	15.6%

170 $\delta/(1 - \varepsilon)^n$ when it uses n real demands, whatever the simulator and however
171 much simulation it uses. Accepting a perfect unit with certainty therefore
172 still needs $n \geq n_0(\varepsilon, \delta)$ real demands. The reason is that the simulated
173 data have the same law under both hypotheses, while an all-success real
174 sample is possible under both. This makes precise, for the success-run test,
175 the principle that external evidence cannot shorten a test whose validity
176 must hold for every external law [19], and the related limit on combining
177 evidence for ultra-high dependability [20]. A paired twin is built from the
178 real events and is therefore *not* an independent simulator; pairing is what
179 Result 3 exploits.

180 *Result 2 (the exact price of relaxing confidence).* If the demonstration may
181 accept a worse unit with probability $\delta' > \delta$ when the simulator is wrong—
182 that is, if one accepts a higher consumer’s risk—the real demands can be
183 reduced by at most

$$c = n_0(\varepsilon, \delta) - n_0(\varepsilon, \delta'),$$

184 and this reduction is attained by a two-tier plan: run the zero-failure test with
185 $n_0(\varepsilon, \delta')$ demands when a simulator test passes and with $n_0(\varepsilon, \delta)$ otherwise.
186 The plan keeps confidence $1 - \delta$ only if a simulator of a worse-than-target
187 unit passes with probability at most π^* : 6.1%, 3.0%, 1.5% and 1.0% for
188 $\delta' = 7.5, 10, 15$ and 20% at $\varepsilon = \delta = 5\%$. This requirement is itself a
189 link between simulator and reality, so Result 2 is best read as the price of
190 lowering the confidence, not as a free credit. The frequently used continuous
191 form $\lfloor \ln(\delta'/\delta)/(-\ln(1 - \varepsilon)) \rfloor$ is only a lower bound on c . Table 3 gives the
192 numbers.

193 Monte-Carlo runs match the closed form within 0.29 percentage points

194 on a 40-point grid. Used as an independent simulator (its events resampled
195 independently of the real draws), the twin event pool passes the simulator
196 test in 73.3% of runs, so the credit is only partly realised (73%), and the false-
197 acceptance probability at $p = \varepsilon$ is δ' up to the pass probability: the credit
198 is bought exactly with the extra error. An off-the-shelf synthetic benchmark
199 (the synthetic part of the BMC benchmark [49]; 30 events long enough for the
200 operating point) never passes at $\varepsilon = 5\%$ (0%): it has too few synthetic events
201 to bound its own miss probability below 5%. At $\varepsilon = 20\%$ it passes in every
202 run and buys the same credit as the twin (supplementary Table S11), which
203 shows that passing the simulator test is no evidence of fidelity. Monte-Carlo
204 standard errors are about 0.1%.

205 *Result 3 (paired-twin bound).* For any pairing of real and twin events, the real
206 miss probability is at most the twin miss probability plus the discordance,

$$p \leq q + D,$$

207 with no assumption on the simulator (the bound splits the real misses into
208 those the twin also misses and those it catches). On a single camera the
209 bound does not save demands: if $q \leq q_U$ holds with confidence $1 - \delta/2$ and
210 $D \leq \varepsilon - q_U$ is demonstrated by a zero-discordance run at level $\delta/2$, the
211 camera needs $\lceil \ln(\delta/2)/\ln(1 - \varepsilon + q_U) \rceil$ paired events, i.e. 72, 91 and 122
212 for $q_U = 0, 1, 2\%$ at $\varepsilon = \delta = 5\%$, all above $n_0 = 59$. A twin built with a
213 pessimistic (shorter) event length remains valid: its miss probability can only
214 grow and its discordance only shrink. On the fixed-camera twin, shortening
215 every event by 50% raises the twin miss rate at OP* from 1.37% to 2.36%.
216 With a uniform phase the discordance probability of an event is $|R' \setminus R|/K$,
217 where R, R' are the residues modulo K of the real and twin hit frames; the
218 worst-phase indicator $\mathbf{1}\{R' \setminus R \neq \emptyset\}$ dominates it, so the bounds of Result 4
219 also hold for the integer worst-phase counts we report.

220 *Result 4 (transfer across scenes).* Let m calibration scenes contribute $N =$
221 $\sum_s n_s$ paired events, X of them discordant, and let J scenes show at least one
222 discordant event. (a) *Population bound (the transfer guarantee).* If the scenes
223 are drawn independently from a scene population (events within a scene
224 may be dependent), the population-average discordance of a new camera is
225 at most $U(J, m)$ with confidence $1 - \delta$, so its expected miss probability is
226 at most its expected twin miss probability plus $U(J, m)$. The bound counts
227 scenes, not events; it needs no independence of events within a scene. Its price

Table 4: Data needed for the per-camera form of Result 4(c) with no discordance ($J = 0$): minimum number of scenes $m_{\min}$, events per scene n at $m = m_{\min}$ and at $m = 2m_{\min}$, total events, and the asymptotic floor $\ln(1/\delta)/(\gamma t)$.

t	γ	$m_{\min}$	n at $m_{\min}$	n at $2m_{\min}$	total at $2m_{\min}$	floor
2.5%	5%	59	183	28, 3,304	2,397	
2.5%	10%	29	158	28, 1,624	1,198	
5%	5%	59	91	14, 1,652	1,198	
5%	10%	29	78	14, 812	599	
10%	5%	59	44	7, 826	599	
10%	10%	29	38	7, 406	300	

is the number of scenes: with $J = 0$, $U \leq 10\%$ needs 29 scenes and $U \leq 5\%$ needs 59. With the 48 fixed-camera scenes of this study, $J = 0, 1, 2, 3$ give $U(J, m) = 6.1\%, 9.5\%, 12.5\%$ and 15.4% at level δ and $7.4\%, 11.1\%, 14.3\%$ and 17.2% at level $\delta/2$: a single discordant scene costs about three percentage points, and a bound below $\varepsilon/2 = 10\%$ tolerates one discordant scene at level δ and none at $\delta/2$. Adding events to existing scenes does not tighten it; only new scenes do. (b) *Fleet bound (calibrated cameras, independent events)*. For the calibrated cameras, the event-weighted discordance is at most $U(X, N)$ with confidence $1 - \delta$ (for $\delta \leq 1/4$), however heterogeneous the cameras are; hence their event-weighted miss rate is at most their twin miss rate plus $U(X, N)$. Applying it requires a separate bound on the twin miss rate, with its own share of δ ; where real misses exist, direct Clopper–Pearson on them is at least as informative. The proof compares the Poisson-binomial count with a binomial of the same mean [50] and uses a lower bound on the probability that a binomial reaches its mean [51], applied to the complementary count. This bound assumes that events are independent given the scenes, so positive dependence within a scene makes it optimistic, and it says nothing about a new camera. (c) *Per-camera limit*. A per-camera statement, “ $p \leq q + t$ except on a set of cameras of mass γ ”, follows if every scene has at least $n_{\min}$ events, with $\gamma = U(J, m)/(1 - (1 - t)^{n_{\min}})$. Table 4 turns this into a data requirement: for $\gamma = 5\%$ at least 59 scenes, each with tens to hundreds of events. No public benchmark we know offers this, so the per-camera form is reported as a limit.

Demonstration procedure. The procedure evaluated in Section 6 is: (1) declare the domain and pair every real event of m calibration scenes from it with its twin; count discordant scenes J ; (2) *domain rule*: if $U_{\delta}(J, m) > \varepsilon/2$, withdraw the transfer claim for the whole domain (a design rule for go/no-go,

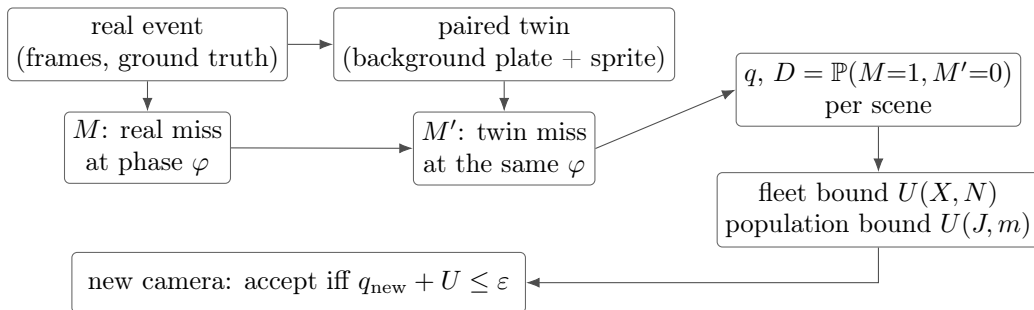


Figure 1: The demonstration with a paired twin. Each real event is paired with a synthetic twin that replaces the object and keeps the scene; both are evaluated by the same detector at the same sampling phase. Discordances calibrated across scenes bound the miss probability of a new camera through $p \leq q + D$.

not a theorem); (3) for a new camera of the declared domain, run its twin until its own miss probability is bounded, q_U , at level $\delta/2$ (a twin *run* is one rendering of one synthetic event in one sprite variant; runs are treated as independent, which is optimistic when runs share a trajectory); (4) accept the camera with no real demand if $q_U + U_{\delta/2}(J, m) \leq \varepsilon$, otherwise fall back to the zero-failure test (Fig. 1). By Result 4(a) and a union bound, with confidence $1 - \delta$ the population-average discordance is at most $U_{\delta/2}(J, m)$. Among *accepted* cameras the average discordance can be larger, by at most the factor $1/\mathbb{P}(\text{accept})$, because acceptance selects cameras with a low twin miss rate; the average miss probability of accepted cameras is therefore at most $\varepsilon - U_{\delta/2} + U_{\delta/2}/\mathbb{P}(\text{accept})$. At OP*, where 95.8% of the static cameras are accepted, the inflation is 0.3 percentage points; out of domain it is unbounded (Section 6.3). The acceptance is *not* a per-camera guarantee. In the audit we nevertheless count, per camera, acceptances with $p_s > \varepsilon$ (“false acceptances”) as a diagnostic of how the average guarantee behaves on individual cameras.

5. Experimental protocol

Data. The corpus is rebuilt from the raw detections and pixel ground truth of CDnet 2014 [52] and LASIESTA [53] (frames in true temporal order), with the synthetic part of the BMC benchmark [49] as an off-the-shelf simulator; the detector-defined corpus has 3,170 events from 101 real videos. For the twin we use ground-truth events: a frame is a *hit* if a detector box covers at

277 least half of the largest ground-truth component or overlaps its bounding box
 278 with $\text{IoU} \geq 0.3$, and an event counts if its largest object has at least 256 pixels
 279 (primary definition; the original definition keeps every ground-truth event).
 280 Scenes are assigned to tiers by the category labels of the two datasets. The
 281 *static* (main) domain is every CDnet calibration scene outside the categories
 282 *PTZ*, *cameraJitter*, *nightVideos* and *turbulence*, and every LASIESTA cali-
 283 bration scene outside the categories SM (simulated camera motion) and MC
 284 (moving camera): 48 scenes (28 CDnet, 20 LASIESTA; 161 events of any
 285 duration, 114 of at least 32 frames). The other tiers are night (6 scenes),
 286 turbulence (4 scenes; only 2 were run and they hold 2 events, so the tier is
 287 reported but not certified) and camera jitter (26 scenes: CDnet *cameraJitter*
 288 and LASIESTA SM); the 4 pan-tilt-zoom and 4 moving-camera scenes are
 289 excluded, because the twin assumes a static background. This definition was
 290 fixed after the initial version of this study, whose static domain was wider; no
 291 twin or detector was re-run, and the supplement compares the static domain
 292 of the three versions (Table S16).

293 *Detector and twin.* The detector is a fixed single-stage segmentation network
 294 (YOLO family [54]; yolo26s-seg [55], confidence 0.25, input 640 px, CPU).
 295 Per scene, a background plate is the per-pixel median of up to 60 frames
 296 without ground-truth foreground. For each real event and each of three sprite
 297 variants (fixed seeds), the twin removes the real object (mask dilated by 3
 298 px), fills it from the plate, and draws a synthetic object on the ground-truth
 299 trajectory: a vehicle cut from Virtual KITTI 2 [56], chosen by aspect ratio,
 300 or a procedurally dressed human rendered with MPFB [57] with a four-frame
 301 walking cycle mirrored to the motion direction. The sprite height equals the
 302 ground-truth box height, its width is clamped to $\pm 10\%$ of the box width, its
 303 brightness follows the frame luminance, and a soft shadow is added. Only the
 304 object is synthetic; scene, illumination and background clutter are real. The
 305 detector runs on twin frames until the twin hits cover all residues modulo 48,
 306 which is exact for every K dividing 48. The median detector confidence on
 307 twin hits is 0.78 (real: 0.87). Building the twins of the 48 static scenes took
 308 14.1 CPU-hours (median 0.20 per scene) on a laptop CPU.

309 *What the twin consumes.* The paired twin uses, per camera, empty frames
 310 (for the plate) and the annotated trajectories of the camera’s real events (for
 311 sprite placement). Calibration therefore costs annotated events: 114 at OP*,
 312 i.e. 2.4 per static camera when amortized over the domain. For a *new* camera,

the exchange rates below assume that the twin miss rate q_{new} can be measured without that camera’s real events, e.g. with synthetic trajectories drawn on its empty background. We did not build such trajectory-free twins; we use the paired q_s in their place. The discordance of a trajectory-free twin has not been calibrated, so the reported savings are conditional on the paired-twin discordance carrying over, and are upper bounds.

Operating points. An operating point is $(\varepsilon, d_{\min}, K)$ with events of at least $d_{\min}$ frames. The primary point $\text{OP}^* = (20\%, 32, 16)$ at $\delta = 5\%$ was fixed by a rule chosen before the twin was built (the largest K for which direct certification needs at most 150 events at 80% power and the 44 calibration scenes available at that time give a population bound of at most $\varepsilon/2$), using the pooled real miss rate of all CDnet events of every category (11.3% over 161 such events, 112 events for direct certification). In the static domain the real miss rate at OP^* is only 0.88%; we report this, keep OP^* , and add the sensitivity points OP-A (20%, 32, 8) and OP-C (10%, 32, 1) and the full grid $d_{\min} \in \{1, 4, 8, 12, 16, 24, 32\}$, $K \in \{1, 2, 3, 4, 6, 8, 12, 16, 24, 48\}$. The night and turbulence tuning criteria and their held-out scenes were fixed before tuning. These choices are dated in internal project records but were not publicly registered. Six choices were made after results were available and are disclosed: (i) the primary event definition (largest object at least 256 pixels) was adopted after the initial twin run, with the original definition kept as a co-reported sensitivity (Table 11); (ii) the list of LASIESTA calibration scenes, before the split into tiers, was rebuilt (20 to 48 scenes) after correcting the frame order of that dataset, before the twin was built; (iii) the two held-out turbulence scenes were not run; (iv) the CDnet *cameraJitter* scenes and (v) the LASIESTA SM and MC scenes were removed from the static domain after static-domain results were available; and (vi) a second operating point with more real misses was examined after inspecting the data (Section 6.3). The operating point OP^* , the audit design and the night criterion were fixed before the static-domain twin results; everything else in Section 6 is exploratory.

Why a paired twin. A reweighting alternative [58] needs overlap between synthetic and real events. A classifier separating BMC events from real events reaches a median AUC of 0.64 with temporal features, 0.94 with geometry and 0.77 with detector confidence alone (all targets); the gap lies mostly in geometry, so a reweighted synthetic benchmark cannot stand in for the real event distribution, and the twin is matched on geometry by construction.

Table 5: Certificates by tier at OP* ($\delta = 5\%$). Discordant events are counted at the worst sampling phase (in parentheses: phase-averaged count, mean over the three sprite variants; per-scene tables use the majority rule, which gives 8.31 at night); $\hat{p}$, $\hat{q}$: real and twin miss rates averaged over the phases / at the worst phase (any fixed deployment phase); J = scenes with a discordant event.

tier	events	$\hat{p}$ avg/worst	$\hat{q}$ avg/worst	discordant	$U(X, N)$	J/m	$U(J, m)$
static (main)	114	0.9%/0.9%	1.4%/3.5%	0 (0.0000)	2.59%	0/48	6.05%
camera jitter	62	0.5%/4.8%	0.0%/0.0%	3	12.0%	1/26	17.0%
night	25	36.2%/52.0%	4.8%/16.0%	12 (8.21)	65.9%	5/6	99.1%
turbulence	2	6.2%/50.0%	0.0%/0.0%	1	97.5%	1/2	97.5%
all tiers	203	5.2%/8.9%	1.4%/3.9%	16	11.7%	7/82	15.4%

350 *Audit.* Leave-one-scene-out (LOSO): each scene is certified with the bound
351 computed from the other scenes. Monte Carlo (10,000 draws): calibration
352 scenes are resampled from the static domain and a new camera is drawn from
353 it; a false certificate is a certified camera with $p_s > \varepsilon$ (the resampled calibration
354 set may contain the test camera, which makes this check optimistic; the
355 LOSO audit does not have this defect); the “truth” of a camera is its own
356 event pool, exact over phases. Two must-fail scenarios are run (Section 6.3).
357 All randomness uses seed 42.

358 6. Results

359 6.1. Certificates by tier

360 In the static domain, 0 of 114 events at OP* are discordant; hence
361 $U(X, N) = 2.59\%$ and, with $J = 0$ of 48 scenes, $U(J, m) = 6.05\%$ (Ta-
362 ble 5, Fig. 2). Both are below $\varepsilon/2 = 10\%$. In the camera-jitter tier, 3 of 62
363 events are discordant, all in the CDnet scene *traffic*, which has real camera
364 shake; the LASIESTA scenes with simulated camera motion have no discor-
365 dant event. At night, 12 of 25 events are discordant (48%; phase-averaged
366 8.21) and 5 of 6 scenes; pooling all tiers raises the bounds to 11.7% and
367 15.4%.

368 6.2. Exchange rate

369 Fig. 3 and Table 7 show the real events a static camera saves when the
370 twin replaces direct certification, across 70 operating points. A camera is
371 certified with no real event when its twin miss rate plus the transfer bound
372 (both at $\delta/2$, the bound computed without the camera) stays below ε . At

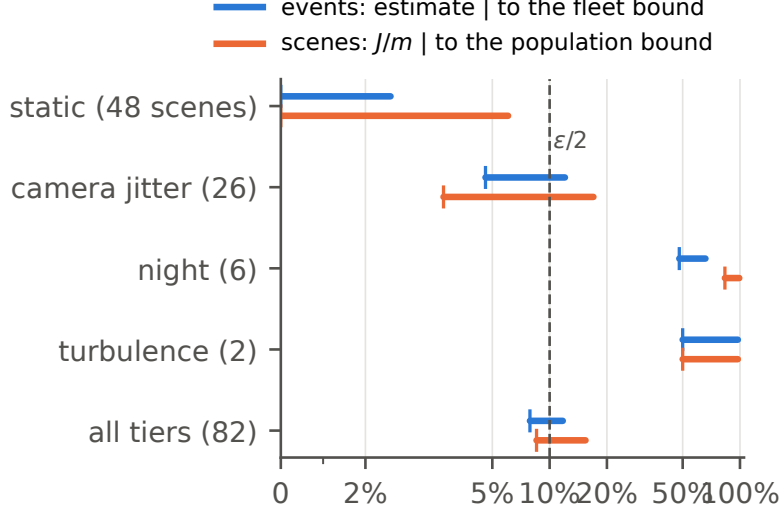


Figure 2: Certificates by tier at OP^* (event unit, worst phase). Blue: discordant events per event and the fleet bound; orange: discordant scenes per scene and the population bound. Dashed: $\varepsilon/2$.

373 OP^* the direct test needs only $n_0 = 14$ events, because the static-domain
 374 miss rate is low; a twin run to convergence saves those 14 events at 46 of 48
 375 cameras, at a measured cost of 0.069 CPU-hours per saved event (conditional
 376 on the discordance carrying over, Section 5; no accepted camera has $p_s > \varepsilon$
 377 at OP^* , but acceptances were not audited at every grid cell). **The twin as**
 378 **built saves no real events; the saving requires about 28 twin runs**
 379 **per camera.** With three runs per event, as built, the twin’s own miss rate
 380 cannot be bounded tightly enough, and at most 4 cameras save anything
 381 at any grid point. The saving grows with the real miss rate, because direct
 382 certification becomes expensive: it reaches hundreds of events per camera at
 383 some grid cells with $\hat{p}$ above 10%, while at $K = 48$ and closer to ε the twin
 384 route saves nothing (Table 7). The target $\varepsilon = 20\%$ is a demonstrator, not
 385 a safety target. At $\varepsilon = 5\%$ the domain rule requires a population bound
 386 of at most $\varepsilon/2 = 2.5\%$, which needs at least 119 calibration scenes even
 387 with no discordant scene; with 48 scenes the twin route is not available at
 388 $\varepsilon = 5\%$, whatever the twin budget. For the calibrated cameras themselves,
 389 direct Clopper–Pearson on their pooled real misses gives 4.1%, close to the

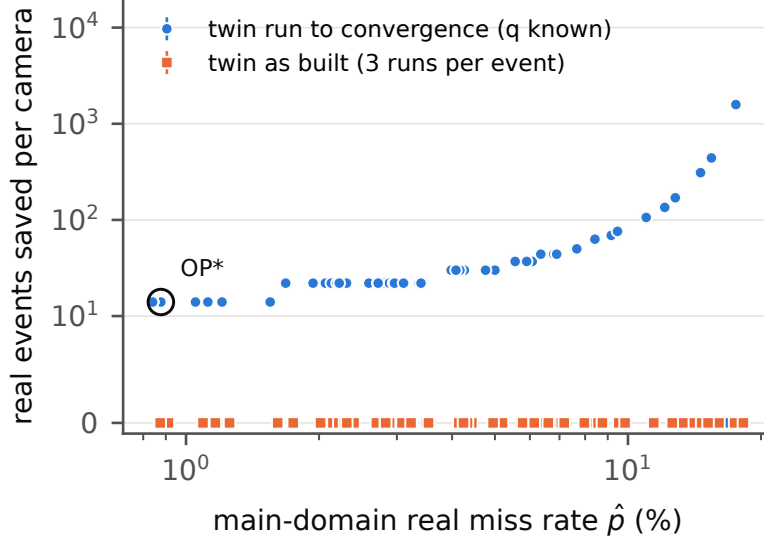


Figure 3: Exchange rate: real events saved per static camera (median and interquartile range over 48 cameras) against the static-domain real miss rate, one point per $(d_{\min}, K)$ at $\varepsilon = 20\%$. Blue: twin run to convergence; orange: twin as built (three runs per event). Vertical axis: symmetric log (linear below 10). Circled: OP*.

390 fleet discordance bound 2.59%: the fleet certificate adds little where real data
 391 exist, and its role is as the building block of the transfer.

392 6.3. Audit and must-fail scenarios

393 *Is the certificate right because it is empty?* Table 8 cross-tabulates real
 394 and twin outcomes. In the static domain at OP* almost every event is caught
 395 by both: 1 events have a real miss at some phase, in 1 scenes, and the
 396 twin also misses 100% of the expected real misses, i.e. one event. The zero-
 397 discordance result is therefore mostly a statement about a detector that rarely
 398 misses in this domain, not a test of the twin. At night the twin sees almost
 399 none of the real misses.

400 *LOSO and Monte Carlo.* At OP*, the number of static scenes whose
 401 discordance exceeds the bound computed without them is 0 of 48 (Fig. 5);
 402 the Monte-Carlo false-certification rate is 0.0%. Because real misses are rare
 403 at OP*, this audit cannot reveal much: only 1 static scene has $p_s > \varepsilon$ there
 404 (the one holding the single missed event), and it is refused, so the 0% rate

Table 6: Exchange rate at three operating points (48 static cameras). n_{dir} : real events of the smallest direct plan (failures allowed, Clopper–Pearson acceptance) with 80% power at the pooled static-domain $\hat{p}$; the zero-failure plan alone needs n_0 events; saved: median real events saved per camera by a twin run to convergence; runs: twin runs needed per camera; CPU-h: CPU-hours per saved real event.

point	$\hat{p}$	n_{dir}	saved	cams	runs	CPU-h
OP* (20%, 32, 16)	0.88%	14	14	46	28	0.069
OP-A (20%, 32, 8)	0.88%	14	14	47	28	0.064
OP-C (10%, 32, 1)	0.88%	46	46	47	149	0.104
worst-phase convention (valid for any fixed phase):						
OP*	0.9%	14	14	44	28	–
twin as built: cameras saving = 3, 3, 0 (median saving 0)						

Table 7: Operating-point grid at $\varepsilon = 20\%$ (condensed; full grid in the supplement). Each cell: static-domain real miss rate $\hat{p}$ / real events for direct certification / median real events saved per camera by a twin run to convergence. The median saving of the twin as built is 0 at every cell. OP* is (32, 16).

$d_{\text{min}} \backslash K$	1	8	16	24	48
32	0.9%/14/14	0.9%/14/14	0.9%/14/14	1.2%/14/14	5.9%/37/0
24	0.8%/14/14	1.1%/14/14	1.7%/22/22	2.3%/22/22	8.2%/57/0
16	1.6%/14/14	2.6%/22/22	4.3%/30/30	6.1%/37/37	13.5%/210/0
12	2.3%/22/22	3.4%/22/22	5.0%/30/30	6.8%/44/44	14.1%/260/0
8	2.2%/22/22	4.0%/30/30	6.9%/44/44	9.2%/69/69	16.8%/901/0
4	2.7%/22/22	8.4%/63/63	12.8%/170/170	15.4%/441/441	23.0%/–/0
1	3.1%/22/22	14.6%/310/310	19.5%/–/0	22.2%/–/0	29.4%/–/0

405 rests on that one scene. A scene above its leave-one-out bound would still be
 406 compatible with a population-average guarantee. The informative tests are
 407 must-fail 2 and the camera-jitter check below, which can fail and do.

408 *Must-fail 1: temporal.* At $K = 48$, 3 static scenes have $p_s > \varepsilon$ and the
 409 procedure refuses them in 100% of runs; but most of these misses are temporal
 410 (events shorter than K , which the twin reproduces by construction); the
 411 number of scenes that exceed ε through detector misses is 1 (Table 9, top).

412 *Must-fail 2: detector.* We add k of the 6 night scenes to the calibration
 413 as if they were static-domain scenes; at OP* every real miss of these scenes is
 414 a detector miss. The transfer bound rises with k (Fig. 4, Table 9, bottom):
 415 from 6.1% to 18.5% at OP*, crossing $\varepsilon/2$ for some subsets at $k = 2$ and for
 416 all at $k = 3$, so the domain-level certificate is withdrawn; no static camera is

Table 8: Real versus twin outcomes (expected event counts under a uniform sampling phase; twin = majority of three variants). Recall: share of expected real misses that the twin also misses.

	static, OP*	night, OP*
real miss, twin miss	1.00	0.75
real miss, twin catch (D)	0.00	8.31
real catch, twin miss	0.56	0.44
real catch, twin catch	112.44	15.50
events	114	25
events missed at some phase	1	13
twin recall of real misses	100%	8%

417 ever falsely certified (0). *Out-of-domain pass-through*: the per-camera rule
 418 does not protect the night cameras themselves. The night twin misses almost
 419 nothing (its appearance model only dims the sprite), so each night camera,
 420 judged with a bound computed from the static scenes, is certified; 4 of 6 night
 421 cameras have $p_s > \varepsilon$ and are falsely certified at OP* (Table 9, middle). Each
 422 night miss rate rests on 2 to 10 events and is uncertain; the finding is that the
 423 acceptance rule admitted these cameras although their observed miss rates
 424 exceed ε and their twins miss almost nothing, not a precise estimate of their
 425 miss rates.

426 *Camera jitter*. The same pass-through occurs for a shaking camera, but
 427 it rests on a single scene. Judged by the twin route calibrated on the static
 428 scenes (worst-phase convention, Section 6.4), 26 of the 26 camera-jitter cam-
 429 eras are accepted and 1 of them is falsely accepted: *traffic*, a CDnet scene with
 430 real camera shake, whose worst-phase real miss rate is 37.5% (per-camera ta-
 431 ble in the supplement). Its twin catches every event at every phase, because
 432 the synthetic object is drawn on a stabilised plate while the real frames shake;
 433 under the phase-averaged reading this camera stays below ε (Table 9, camera-
 434 jitter rows), so this failure appears only for a fixed deployment phase. The
 435 LASIESTA scenes with simulated camera motion produce no discordance
 436 and no false acceptance, so they add little evidence either way about real
 437 shake. Under the same worst-phase rule 4 of 6 night cameras are falsely
 438 accepted at OP*.

439 *A post-hoc operating point*. To see the certificate under more real misses,
 440 the point $(d_{\min}, K) = (16, 16)$ was examined after inspecting the data; it
 441 fails the criterion we fixed before recomputing it on the static domain (both
 442 bounds at most $\varepsilon/2$ and no false acceptance), because its population bound
 443 is 12.5%, and it is not reported further.

Table 9: Must-fail scenarios. Top: temporal ($d_{\min} = 1$, $K = 48$), static scenes with $p_s > \varepsilon$, real misses split into detector and temporal misses; all refused in every Monte-Carlo run. Middle: night scenes at OP* (every miss is a detector miss), and whether each is falsely certified when judged alone (phase-averaged); then the same for the camera-jitter scenes (CDnet scenes with real shake listed; the LASIESTA scenes with simulated camera motion pooled, event-weighted). Bottom: population bound when k night scenes leak into the calibration set (median over subsets; withdrawn = share of subsets with a bound above $\varepsilon/2$).

<i>1: temporal</i>						
scene	n	p_s	detector	temporal	q_s	refused
port_0_17fps	5	99.6%	43.3%	56.2%	100.0%	100%
tramCrossroad_1fps	11	77.7%	6.1%	71.6%	74.2%	100%
tunnelExit_0_35fps	34	83.6%	5.2%	78.4%	87.9%	100%
<i>2: detector (night)</i>						
scene	n	p_s	q_s	D_s	disc.	false cert.
bridgeEntry	2	15.6%	0.0%	15.6%	1	no
busyBoulevard	4	96.9%	0.0%	96.9%	4	yes
fluidHighway	4	26.6%	0.0%	26.6%	2	yes
streetCornerAtNight	10	28.1%	11.9%	20.6%	4	yes
tramStation	2	0.0%	0.0%	0.0%	0	no
winterStreet	3	33.3%	0.0%	33.3%	1	yes
<i>3: detector (camera jitter)</i>						
boulevard	8	0.0%	0.0%	0.0%	0	no
traffic	8	3.9%	0.0%	3.9%	3	no
LASIESTA SM (24 scenes)	46	0.0%	0.0%	0.0%	0	0 of 24
k leaked	0	1	2	3	4	6
U_{pop}	6.1%	9.3%	12.1%	13.2%	14.2%	18.5%
withdrawn	0%	0%	67%	100%	100%	100%

6.4. Fixed-phase and site-level checks

Worst phase. A deployed camera keeps one sampling phase. Counting an event as missed if it is missed at any phase, for the real event, the twin (majority of variants) and the discordance alike, keeps the bound valid for every fixed phase, because an event missed at some phase is either missed by the twin or discordant at that phase. At OP* the static-domain real and twin miss rates become 0.9% and 3.5% (phase-averaged 0.9% and 1.4%), while the discordance counts and both bounds are unchanged (Table 5). The direct plan then needs 14 events, and a twin run to convergence accepts 44 of 48 cameras, saving 14 events each (Table 6); the number of accepted static cameras with a worst-phase real miss rate above ε is 0.

Sites. Scenes filmed at the same site share background, viewpoint and

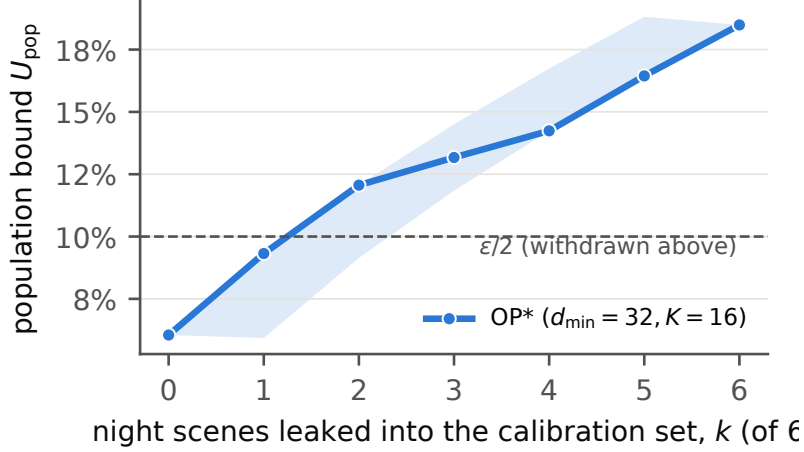


Figure 4: Must-fail 2: the population bound when k night scenes leak into the static calibration set (median over the $\binom{6}{k}$ subsets, band: min-max). The domain certificate is withdrawn above $\varepsilon/2$. Individual night cameras are still certified when judged alone (out-of-domain pass-through, text).

illumination, so the scene may be too fine a unit of independence. Grouping the static scenes into 17 sites (CDnet challenge categories and LASIESTA sequence groups, a coarse proxy for sites), the number of sites with a discordant event at OP^* is 0, yet with so few units the population bound is $U(J_{site}, m_{site}) = 16.2\%$, above $\varepsilon/2$: under the domain rule the transfer claim is withdrawn at site level. The leave-one-site-out audit finds 0 sites above their bound at OP^* . The transfer certificate therefore stands at scene level and is withdrawn at site level; it holds only if scenes, not sites, are the independent units.

6.5. Failure modes

At night the twin sees few real misses: it predicts 8% of the expected missed events, and the tier bound is 99.1%. Fig. 6 shows why: the appearance model scales sprite brightness with the scene, but has no headlights, glare or sensor noise. An attempt to fix this was specified in advance: night-specific appearance profiles were tuned on four night scenes and judged on two held-out scenes, with success defined as a median held-out per-scene discordance of at most 5%. The best profile barely moved the tuning discordance (47.2% to 47.0%, phase-averaged) and the held-out median was 23.2% on 14 events,

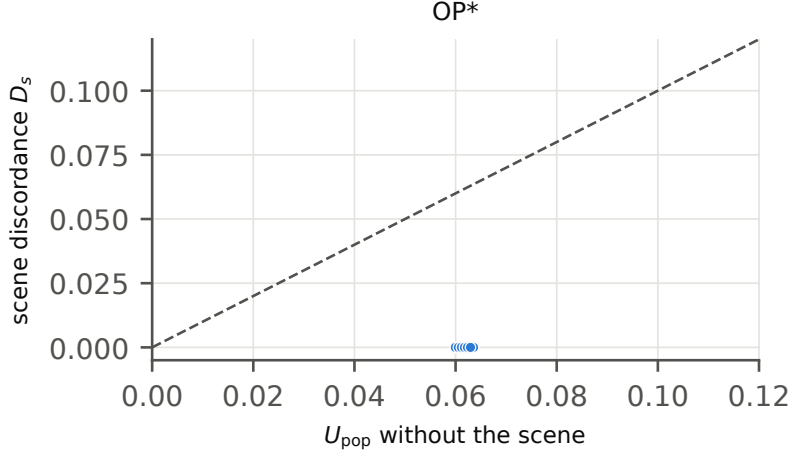


Figure 5: Leave-one-scene-out: per-scene discordance against the population bound computed without the scene. Points above the diagonal exceed the bound; the proposition bounds the population mean, not each scene.

so the criterion failed, tuning was stopped, and night is declared outside the domain. Camera jitter is the second named failure mode, observed in one scene with real shake: the twin is drawn on a stabilised background plate, so it cannot reproduce the misses caused by camera shake (over the tier, twin worst-phase miss rate 0.0% against a real 4.8% at OP*; Table 5). One scene does not show how often real shake produces this failure. The turbulence tier holds 2 events at OP* and is not certified.

6.6. Sensitivity

Event definition. Keeping every ground-truth event (222 instead of 161 static events of any duration) leaves OP* almost unchanged: 116 events, 0 discordant, $U(X, N) = 2.55\%$, $J = 0$ (Table 11). *Operating point.* At OP-A and OP-C the static-domain miss rate is 0.88% and 0.88%, and no static-domain event is discordant. *Twin details.* The majority rule over sprite variants (1, 2 or 3 of 3) does not change any static-domain count. Table 10 compares twin and real hit rates on the same frames: the twin is pessimistic at every aspect error except one vehicle bin, and both hit rates fall when the ground-truth box has an unusual aspect (occlusion, objects entering the frame). Clamping the sprite width, which affects 70% of twin frames (more than the frames with an aspect error above 10% in Table 10, because 1-pixel



Figure 6: Synthetic person sprites (CC0) on a neutral background under the brightness gain used for day and for night scenes. Dimming alone does not reproduce night imaging, which explains the low twin miss rate at night.

rounding also triggers the clamp), lowers the twin hit rate by only 3.0 points more than the real hit rate on the same frames (631 event-variant pairs): it inflates q , not D . The median detector confidence on hits is lower for the twin than for real objects, especially for persons. Full tables are in the supplement.

7. Discussion and limitations

A narrow window. Simulation pays in events only when direct certification is expensive, that is, when the real miss rate is not far below ε ; there the twin must also be accurate, and discordances make the transfer bound grow. When the miss rate is low, n_0 is small and there is little to save. The grid (Table 7) shows both ends: 14 events at OP*, hundreds at some operating points with $\hat{p}$ above 10%, and nothing where $\hat{p}$ comes close to ε .

Time. The waiting time saved at camera s is $n_{\text{saved}}/\lambda_s$, where λ_s is the event rate in deployment; we do not estimate it, because the benchmarks publish no frame rates and their clips are dense in events. If one CPU-hour is valued like one hour of waiting, the twin breaks even when $\lambda_s < 1/c \approx 14$ events per hour at OP*, with c the CPU-hours per saved event; this equivalence is an assumption, not a measurement.

Domain. The transfer bound is a population-average guarantee: it withdraws the domain certificate when out-of-domain scenes enter the calibration, but it does not detect an out-of-domain camera. The domain must therefore

Table 10: Twin fidelity on the same frames (static domain): hit rates by object kind and sprite aspect error before the width clamp, and median detector confidence on hits. Negative differences make the twin pessimistic.

kind	aspect error	frames	twin hit	real hit	diff. (pts)
person	$\leq 10\%$	8,870	91.7%	94.8%	-3.2
person	10-20%	5,899	90.1%	94.2%	-4.2
person	20-40%	4,404	74.0%	86.2%	-12.2
person	$> 40\%$	5,449	34.4%	44.5%	-10.0
vehicle	$\leq 10\%$	4,105	86.9%	89.0%	-2.1
vehicle	10-20%	2,569	88.0%	90.2%	-2.2
vehicle	20-40%	3,087	87.6%	85.1%	+2.5
vehicle	$> 40\%$	3,881	64.9%	74.2%	-9.3
person	confidence		0.82	0.90	
vehicle	confidence		0.72	0.78	

Table 11: Event-definition sensitivity (static domain). Primary: largest object ≥ 256 px; original: every ground-truth event. Discordant events at the worst phase; LOSO: scenes above the bound computed without them.

definition	point	N	$\hat{p}$	disc.	$U(X, N)$	$J / U(J, m)$	LOSO
primary	OP*	114	0.88%	0	2.59%	0 / 6.1%	0
	OP-A	114	0.88%	0	2.59%	0 / 6.1%	0
	OP-C	114	0.88%	0	2.59%	0 / 6.1%	0
original	OP*	116	0.86%	0	2.55%	0 / 6.1%	0
	OP-A	116	0.86%	0	2.55%	0 / 6.1%	0
	OP-C	116	0.86%	0	2.55%	0 / 6.1%	0

514 be declared and checked by other means (for example, illumination statis-
515 tics), and a per-camera guarantee requires the data of Result 4(c). Three
516 items are left for future work: a twin that needs no annotated real trajec-
517 tory of the new camera, with its own calibrated discordance; a pre-specified
518 in-domain stress test with cameras whose real miss rate exceeds ε ; and a
519 domain check (for example a per-camera discordance sentinel on a few real
520 events) to stop out-of-domain pass-through.

521 *Data and twin.* No static scene has 59 or more qualifying events, so per-
522 camera ground truth is the scene’s own finite event pool; the twin builds its
523 plate from real empty frames and therefore needs a static background; the
524 sprite library lacks elevated viewpoints for vehicles; and we did not use a full
525 3-D simulator. The operating point was fixed from the pooled miss rate of all
526 CDnet events, which turned out to be much higher than the static-domain
527 rate; we kept it rather than re-select after seeing the twin results. Scenes are
528 not independent draws: several LASIESTA sequences share a room or set-

up, and the site-level bound of Section 6.4 exceeds $\varepsilon/2$. The cost comparison counts real events and CPU-hours only; annotating the calibration events and building plates are additional costs. Finally, an event miss is a component-level failure; linking it to a system hazard requires a hazard analysis of the application.

Independence and phase. The results assume independent events given the scene and independent scenes. Events of one camera share a background, a detector and, in deployment, a single sampling phase; benchmark scenes are not a random sample of deployments. Correlation of events within a scene does not invalidate the population bound of Result 4(a) or the leave-one-scene-out audit, because both count scenes, not events: a scene contributes one indicator whatever the dependence among its events. It does affect the fleet bound of Result 4(b), which counts events and becomes optimistic under positive within-scene dependence; the population bound needs only independence across scenes. The main results certify the phase-averaged miss rate; the worst-phase convention of Section 6.4 covers any fixed phase at a higher real-event cost.

Threats to validity. The audit tables and figures use level δ , while acceptance uses $\delta/2$ for the transfer term; at $\delta/2$ the bounds are larger (Result 4(a)). Software: Python 3.11.9, NumPy 1.26.4, SciPy 1.16.3, pandas 2.3.3, Ultralytics 8.4.7, OpenCV 4.10.0; CPU: Intel Core i7-1185G7. The detector, the event definition and the hit rule are fixed; a different detector changes p , q and D but not the theorems. The number of scenes, not the number of events, limits the transfer certificate: the static domain reaches $U(J, m) = 6.05\%$ with no discordant scene among 48, and more scenes are needed for tighter bounds. The audit at OP^* is weak because real misses are rare there. The label-based static domain was fixed after the initial version of this study (Section 5); its clean audit is therefore partly a result of that post-hoc choice.

Practical guidance. The results suggest a simple decision rule for practitioners who consider replacing a field demonstration with a twin. (i) Compute n_0 at the target (ε, δ) and estimate the real miss rate; if it is far below ε , the direct test is already short and a twin can save at most n_0 events. (ii) If a twin is built, declare its domain before calibration and calibrate discordance on as many scenes of that domain as possible; the number of scenes sets the resolution (Result 4(a)). (iii) Run each new camera’s twin long enough to bound its own miss rate; three runs per event, as here, are not enough. (iv) Do not use the transfer certificate to admit a camera whose conditions dif-

567 fer from the calibration domain (night, a shaking mount): the certificate is
568 withdrawn when such scenes enter the calibration, but a single such camera
569 passes through.

570 8. Conclusion

571 Without an assumption linking simulation to reality, synthetic evidence
572 buys no real events in zero-failure certification, and relaxing the error level
573 buys an exactly computable number. A paired twin turns the question into
574 one of discordance. The population bound is the transfer guarantee: it counts
575 scenes, tolerates dependence within a scene, and its price is the number of
576 scenes; the fleet bound covers only the calibrated cameras and assumes inde-
577 pendent events. On the 48 static scenes of this study the transfer certificate
578 stands at scene level ($U(J, m) = 6.05\%$) but is withdrawn at site level, where
579 17 sites give 16.2%, above $\varepsilon/2$. Measured on public benchmarks, the ex-
580 change rate is modest and domain-bound; at $\varepsilon = 5\%$ the transfer needs at
581 least 119 calibration scenes. The certificate has two named failure modes,
582 night and camera jitter (the latter seen in one scene with real shake), in
583 which out-of-domain cameras pass through unless the domain is declared.

584 CRediT authorship contribution statement

585 **Lam Quoc Dat:** Conceptualization, Methodology, Software, Validation,
586 Formal analysis, Investigation, Data curation, Writing – original draft, Writ-
587 ing – review & editing, Visualization.

588 Declaration of competing interest

589 The author declares no competing interests. This research received no
590 specific grant from any funding agency in the public, commercial, or not-for-
591 profit sectors.

592 Declaration of generative AI and AI-assisted technologies in the 593 manuscript preparation process

594 During the preparation of this work the author used an AI assistant
595 (Claude, Anthropic) to write analysis and plotting scripts, to check the num-
596 bers of the manuscript against the result files, and to draft and edit text.
597 After using this tool, the author reviewed and edited the content as needed
598 and takes full responsibility for the content of the publication.

599 Data availability

600 Code, event manifests, result tables and the CC0 person-sprite set are
 601 publicly available at <https://github.com/lamquocdat01/no-free-credit-ress-artifact> (tag `ress-v1.2`, commit 890465bd9267). Raw video frames
 602 (CDnet 2014, LASIESTA, BMC) and Virtual KITTI 2 sprites are not redis-
 603 tributed under their licences; the repository provides download instructions
 604 and SHA-256 checksums to regenerate every derived file.
 605

606 Appendix A. Formal statements and proofs

607 This appendix states the results of Section 4 formally and proves them.
 608 Theorem A.1 is Result 1, Theorem A.3 is Result 2, Lemma A.5 and Propo-
 609 sition A.15 are Result 3, and Proposition A.11, Theorem A.9 and Proposi-
 610 tion A.13 are Result 4(a), (b) and (c).

611 Appendix A.1. Setting and notation

612 **Events and misses.** A camera (scene) s produces events $i = 1, 2, \dots$ (a
 613 maximal run of frames containing the target object). The system samples
 614 frames periodically with period K and an unknown phase $\varphi \in \{0, \dots, K-1\}$;
 615 a detector is run on the sampled frames. An event is *missed* ($M = 1$)
 616 when no sampled frame of the event is a hit (the detector finds the object).
 617 With $H \subseteq \mathbb{Z}$ the hit frames of the event and φ uniform, $\mathbb{P}(M = 1 \mid H) =$
 618 $1 - |H \bmod K|/K$, where $H \bmod K$ is the set of residues of H . The miss
 619 rate of camera s is $p_s = \mathbb{P}(M = 1)$ over its events and phases. Target: certify
 620 $p_s \leq \varepsilon$ with confidence $1 - \delta$.

621 **Clopper–Pearson bound.** For $x \in \{0, \dots, n\}$, $F_{n,p}(x) = \mathbb{P}(\text{Bin}(n, p) \leq x)$
 622 and $U(x, n) = U_\delta(x, n)$ is the one-sided $(1 - \delta)$ upper Clopper–Pearson bound
 623 [59]: the solution u of $F_{n,u}(x) = \delta$ for $x < n$, and 1 for $x = n$. Since $F_{n,p}(x)$
 624 is strictly decreasing in p , $U(x, n) < p \iff F_{n,p}(x) < \delta$, and U is non-
 625 decreasing in x . Validity: $\mathbb{P}(U(\text{Bin}(n, p), n) < p) \leq \delta$ for every p .

626 **Success run.** $n_0(\varepsilon, \delta) = \lceil \ln \delta / \ln(1 - \varepsilon) \rceil$ is the smallest n with $(1 - \varepsilon)^n \leq \delta$
 627 (e.g. $n_0(5\%, 5\%) = 59$, $n_0(20\%, 5\%) = 14$).

628 **Simulator and twin.** Simulator data Y have a law Q on an arbitrary space.
 629 A *twin* is a paired simulator: every real event i has a synthetic counterpart
 630 with miss indicator M'_i under the same sampling phase. $q = \mathbb{P}(M' = 1)$, and
 631 the *discordance* is $D = \mathbb{P}(M = 1, M' = 0)$ (real miss, twin catch).

632 *Appendix A.2. C1: no free credit*

633 A certification procedure is a (possibly randomised) test $\phi(X, Y) \in \{0, 1\}$
 634 with real data $X = (M_1, \dots, M_n) \sim \text{Bern}(p)^{\otimes n}$ independent of simula-
 635 tor data $Y \sim Q$. It is *valid at level δ on a class $\mathcal{P}$* of pairs (p, Q) if
 636 $\sup_{(p, Q) \in \mathcal{P}, p > \varepsilon} \mathbb{P}_{p, Q}(\phi = 1) \leq \delta$.

637 **Theorem A.1** (C1). *Assume $\mathcal{P}$ places no restriction linking Q and p : for*
 638 *every Q there are $(0, Q) \in \mathcal{P}$ and $(p, Q) \in \mathcal{P}$ for all $p \in (\varepsilon, \varepsilon + \eta)$ for some*
 639 *$\eta > 0$. If ϕ is valid at level δ on $\mathcal{P}$, then for every Q and every simulator*
 640 *sample size,*

$$(1 - \varepsilon)^n \mathbb{P}_{0, Q}(\phi = 1) \leq \delta.$$

641 *In particular, certifying a perfect system ($p = 0$) with probability one requires*
 642 *$n \geq n_0(\varepsilon, \delta)$ real events, whatever the amount of simulator data; power $1 - \beta$*
 643 *at $p = 0$ requires $n \geq \ln(\delta/(1 - \beta))/\ln(1 - \varepsilon)$.*

644 *Proof.* Fix Q and write $g(y) = \mathbb{E}[\phi(0^n, y)]$ (expectation over the internal
 645 randomisation). Under $p = 0$, $X = 0^n$ almost surely, so $\mathbb{P}_{0, Q}(\phi = 1) =$
 646 $\mathbb{E}_Q[g(Y)]$. Under $p > \varepsilon$, the event $\{X = 0^n\}$ has probability $(1 - p)^n$ and is
 647 independent of Y , hence $\mathbb{P}_{p, Q}(\phi = 1) \geq (1 - p)^n \mathbb{E}_Q[g(Y)] = (1 - p)^n \mathbb{P}_{0, Q}(\phi =$
 648 $1)$. Validity gives $(1 - p)^n \mathbb{P}_{0, Q}(\phi = 1) \leq \delta$ for all $p \in (\varepsilon, \varepsilon + \eta)$; letting $p \downarrow \varepsilon$
 649 proves the inequality. With $\mathbb{P}_{0, Q}(\phi = 1) = 1$ it reads $(1 - \varepsilon)^n \leq \delta$, i.e. $n \geq n_0$;
 650 with $\mathbb{P}_{0, Q}(\phi = 1) \geq 1 - \beta$ it gives the power bound. $\square$

651 *Remark A.2.* The argument only uses that the simulator data have the same
 652 law under the two hypotheses and that the all-success real sample has positive
 653 probability under both. It is the success-run analogue of the observation
 654 that external data cannot increase power under strict type-I error control
 655 over all external-data laws [19]. *Check.* The zero-failure test with $n = 59$ at
 656 $p = \varepsilon = 5\%$: $(0.95)^{59} = 0.0485$ (MC 0.0491, 10^5 runs).

657 *Appendix A.3. C2: the price of borrowing*

658 Suppose one accepts a two-tier guarantee: level δ when the simulator is
 659 “right” and only level $\delta' > \delta$ in general.

660 **Theorem A.3** (C2, exact credit). (i) *Converse. Under the assumption of*
 661 *Theorem A.1, any procedure valid at level δ' on $\mathcal{P}$ that certifies a perfect*
 662 *system with probability one uses $n \geq n_0(\varepsilon, \delta')$ real events. Hence the credit,*
 663 *the number of real events saved with respect to $n_0(\varepsilon, \delta)$, is at most*

$$c = n_0(\varepsilon, \delta) - n_0(\varepsilon, \delta').$$

664 (ii) Attainment. Let $T(Y) \in \{0, 1\}$ be any simulator test (“the simulator
 665 passes”) and let B be the procedure: if $T = 1$ run the zero-failure test with
 666 $n_1 = n_0(\varepsilon, \delta')$ real events, otherwise with $n_0 = n_0(\varepsilon, \delta)$. Then B is valid at
 667 level δ' on $\mathcal{P}$, saves exactly c events whenever $T = 1$, and is valid at level δ
 668 on the subclass

$$\mathcal{P}_0 = \left\{ (p, Q) : p > \varepsilon \Rightarrow \mathbb{P}_Q(T = 1) \leq \pi^* \right\}, \quad \pi^* = \frac{\delta - (1 - \varepsilon)^{n_0}}{(1 - \varepsilon)^{n_1} - (1 - \varepsilon)^{n_0}},$$

669 i.e. where a simulator of a system that is truly worse than ε rarely passes.
 670 (iii) Continuous lower bound. $c \geq c^* = \lfloor \ln(\delta'/\delta)/(-\ln(1 - \varepsilon)) \rfloor$.

671 *Proof.* (i) Theorem A.1 at level δ' gives $(1 - \varepsilon)^n \leq \delta'$, whose smallest integer
 672 solution is $n_0(\varepsilon, \delta')$. (ii) For $p > \varepsilon$, with $\pi = \mathbb{P}_Q(T = 1)$ and T independent
 673 of the real data, $\mathbb{P}(B = 1) = \pi(1 - p)^{n_1} + (1 - \pi)(1 - p)^{n_0} \leq (1 - \varepsilon)^{n_1} \leq \delta'$,
 674 and $\leq \delta$ as soon as $\pi \leq \pi^*$ (the expression is affine in π and decreasing in
 675 p). (iii) Write $a = \ln \delta / \ln(1 - \varepsilon)$, $a' = \ln \delta' / \ln(1 - \varepsilon)$ and $r = a - a' =$
 676 $\ln(\delta'/\delta)/(-\ln(1 - \varepsilon))$. Since $n_0(\delta) \geq a$ and $n_0(\delta') < a' + 1$, $c > r - 1$; c being
 677 an integer, $c \geq \lfloor r \rfloor = c^*$. $\square$

678 *Remark A.4* (numbers and checks). At $\varepsilon = \delta = 5\%$: $\delta' = 7.5, 10, 15, 20\%$
 679 give $c = 8, 14, 22, 27$ ($-14\%, -24\%, -37\%, -46\%$ of 59). The exact form is
 680 maximal on all 40 grid points $((1 - \varepsilon)^{n_0(\delta')} \leq \delta' < (1 - \varepsilon)^{n_0(\delta')-1}$: yes); MC
 681 (10^5 runs per point) matches the closed form within 0.29 percentage points.
 682 *Empirical C2* (procedure B , T = the one-sided CP bound of the simulator’s
 683 miss rate $\leq \varepsilon$, 10^5 runs, $\varepsilon = 5\%$, $d_{\min} = 32$, $K = 1$): with the paired twin
 684 of the 48 static scenes (114 events) the simulator passes in 73.3% of runs;
 685 realised credit 5.9/10.3/16.1/19.8 events and false certification at the worst
 686 case $p = \varepsilon$ equal to 6.7%/8.4%/12.2%/15.6% for $\delta' = 7.5/10/15/20\%$ — the
 687 credit is bought exactly with the extra error $\delta' - \delta$. The off-the-shelf synthetic
 688 benchmark (30 events of length ≥ 32) never passes (0.0%): too few synthetic
 689 events to bound its own miss rate below 5%.

690 *Appendix A.4. C3: the twin lemma*

691 **Lemma A.5** (C3). For any pairing (M, M') of a real event and its twin,
 692 $p \leq q + D$ with $D = \mathbb{P}(M = 1, M' = 0)$. No assumption links the twin to
 693 reality.

694 *Proof.* $p = \mathbb{P}(M = 1, M' = 1) + \mathbb{P}(M = 1, M' = 0) \leq \mathbb{P}(M' = 1) + D =$
 695 $q + D$. $\square$

Corollary A.6 (a per-camera twin does not save events). *If $q \leq q_U$ holds with confidence $1 - \delta/2$ and $D \leq \varepsilon - q_U$ is certified for the same camera by a zero-discordance success run at level $\delta/2$, the number of paired real events is $\lceil \ln(\delta/2)/\ln(1 - (\varepsilon - q_U)) \rceil = 72, 91, 122$ for $q_U = 0, 1, 2\%$ at $\varepsilon = \delta = 5\%$, all larger than $n_0 = 59$. The twin can only help when D is calibrated once, on many scenes, and transferred (C_4).*

Remark A.7 (event unit). With a uniform phase, $x_i = \mathbb{P}_\varphi(M_i = 1, M'_i = 0) = |R'_i \setminus R_i|/K$, where R_i, R'_i are the residue sets of the real and twin hit frames of event i (twin catch decided by the majority of three sprite variants). The *worst-phase* indicator $W_i = \mathbf{1}\{R'_i \setminus R_i \neq \emptyset\}$ satisfies $\mathbb{P}(M_i = 1, M'_i = 0 \mid \text{event}) = x_i \leq W_i$. All certificates below are stated for Bernoulli indicators Z_i ; they hold verbatim with $Z_i = W_i$ and then bound the (larger) worst-phase discordance rate. This integer form is the one used for the reported certificates.

Appendix A.5. fleet bound: event-weighted certificate for the calibrated fleet

Lemma A.8 (mean–median). *For $N \geq 2$ and $0 < p < 1 - 1/N$, $F_{N,p}(\lfloor Np \rfloor) > 1/4$. Consequently, if $\delta \leq 1/4$ and $F_{N,p}(x) < \delta$, then $x \leq Np - 1$.*

Proof. Greenberg and Mohri [51] prove $\mathbb{P}(B \geq \mathbb{E}B) > 1/4$ for $B \sim \text{Bin}(N, r)$ with $r > 1/N$. Apply it to $B = N - \text{Bin}(N, p) \sim \text{Bin}(N, 1 - p)$ with $1 - p > 1/N$: $\mathbb{P}(\text{Bin}(N, p) \leq Np) > 1/4$, and $\{\text{Bin}(N, p) \leq Np\} = \{\text{Bin}(N, p) \leq \lfloor Np \rfloor\}$. For the consequence: $F_{N,p}$ is non-decreasing, so $F_{N,p}(x) < \delta \leq 1/4 < F_{N,p}(\lfloor Np \rfloor)$ forces $x < \lfloor Np \rfloor$, i.e. $x \leq \lfloor Np \rfloor - 1 \leq Np - 1$. Check: $\inf F_{N,p}(\lfloor Np \rfloor)$ over $N = 2..2000$ and a fine grid of p is 0.2500, approached at $N = 2, p \rightarrow 1/2$ (the constant $1/4$ is tight). $\square$

Theorem A.9 (fleet bound). *Let $Z_1, \dots, Z_N$ be independent, $Z_i \sim \text{Bern}(d_i)$ with arbitrary $d_i \in [0, 1]$ (heterogeneous across scenes and events), $X = \sum_i Z_i$ and $\bar{d} = \frac{1}{N} \sum_i d_i$. If $\delta \leq 1/4$ and $\bar{d} < 1 - 1/N$, then*

$$\mathbb{P}(U(X, N) < \bar{d}) \leq \delta.$$

With $d_i = D_{s(i)}$ (the discordance of the scene of event i), $\bar{d} = \bar{D}_w = \sum_s n_s D_s / N$, the event-weighted discordance of the calibrated cameras; combined with Lemma A.5, $\bar{p}_w \leq \bar{q}_w + U(X, N)$ with probability at least $1 - \delta$.

727 *Proof.* If $\bar{d} = 0$ there is nothing to prove. Otherwise let $x^* = \max\{x :$
728 $F_{N,\bar{d}}(x) < \delta\}$ (if the set is empty the failure probability is 0). By monotonicity
729 of U , $\{U(X, N) < \bar{d}\} = \{X \leq x^*\}$. By Lemma A.8, $x^* \leq N\bar{d} - 1$. Hoeffding's
730 Theorem 4 [50]: for a sum S of independent Bernoulli variables with mean
731 $N\bar{d}$ and $0 \leq b \leq N\bar{d} - 1$, $\mathbb{P}(S \leq b) \leq F_{N,\bar{d}}(b)$. With $b = x^*$, $\mathbb{P}(X \leq x^*) \leq$
732 $F_{N,\bar{d}}(x^*) < \delta$. $\square$

733 *Remark A.10* (what it does not say; checks). The guarantee is conditional on
734 the calibrated scenes and weighted by events: it does not bound any single
735 camera and does not transfer to a new camera (if scenes are random, X is
736 over-dispersed; e.g. $D_s \in \{0, 1\}$ gives $\mathbb{P}(X = 0) = (1 - \mu)^m \gg (1 - \mu)^N$).
737 *Checks.* Six synthetic heterogeneity patterns on a fixed configuration of 44
738 scenes with $N = 304$ events (a calibration set from an earlier stage of the
739 study, used only as a test bed for the theorem): exact Poisson–binomial
740 failure ≤ 0.0471 , MC (10^5) ≤ 0.0473 . Measured discordance probabilities
741 of the study (event level, main domain, five operating points, as measured
742 and rescaled to mean 2% and 5%; 12 cases): exact failure ≤ 0.0272 , MC
743 ≤ 0.0267 , all $\leq \delta = 0.05$.

744 *Appendix A.6. population bound: transfer to a new camera*

745 **Proposition A.11** (population bound). *Let scenes $S_1, \dots, S_m$ be i.i.d. from*
746 *a scene population, each with $n_s \geq 1$ paired events whose discordance indica-*
747 *tors are independent $\text{Bern}(D_{S_s})$ given the scene, $\mu = \mathbb{E}[D_S]$, and $J = \#\{s :$
748 $X_s \geq 1\}$ with X_s the number of discordant events in scene s . Then*

$$\mathbb{P}(U(J, m) < \mu) \leq \delta.$$

749 *For a new camera drawn from the same population, $\mathbb{E}[p_{\text{new}}] \leq \mathbb{E}[q_{\text{new}}] +$*
750 *$U(J, m)$ with probability $\geq 1 - \delta$ over the calibration, and $\mathbb{P}(D_{\text{new}} > t) \leq$*
751 *$U(J, m)/t$ (Markov).*

752 *Proof.* Let $B_s = \mathbf{1}\{X_s \geq 1\}$. Given the scene, $\mathbb{P}(B_s = 1 \mid S_s) = 1 -$
753 $(1 - D_{S_s})^{n_s} \geq D_{S_s}$ because $(1 - D)^n \leq 1 - D$ for $n \geq 1$. Hence $\beta_s :=$
754 $\mathbb{P}(B_s = 1) \geq \mu$, and the B_s are independent across scenes. By a coupling
755 $(B_s = \mathbf{1}\{V_s \leq \beta_s\} \geq \mathbf{1}\{V_s \leq \mu\}$ with i.i.d. uniforms V_s), J stochastically
756 dominates $\text{Bin}(m, \mu)$. Since $U(\cdot, m)$ is non-decreasing, $\mathbb{P}(U(J, m) < \mu) \leq$
757 $\mathbb{P}(U(\text{Bin}(m, \mu), m) < \mu) \leq \delta$ by Clopper–Pearson validity. The corollary
758 follows from Lemma A.5 applied to the new camera and averaged over the
759 population. $\square$

760 *Remark A.12* (price and checks). The resolution is set by the number of
761 scenes: with $J = 0$, $U \leq 10\%$ needs $m \geq 29$ and $U \leq 5\%$ needs $m \geq 59$.
762 *Checks*. The tight case $D_S \in \{0, 1\}$, $n_s = 1$, $m = 48$ (then $J \sim \text{Bin}(m, \mu)$
763 exactly), $\mu \in \{1, 2, 5, 10, 20\}\%$, 10^5 runs: maximum failure 0.0413. Leave-
764 one-scene-out on the 48 static scenes at the primary operating point ($d_{\min} =$
765 32, $K = 16$): $D_s > U(J_{-s}, m - 1)$ for 0 of 48 scenes (the proposition bounds
766 the population mean, not each scene).

767 *Appendix A.7. per-camera form: a per-camera (quantile) certificate and its*
768 *data requirement*

769 **Proposition A.13** (per-camera form). *Under the assumptions of Proposi-*
770 *tion A.11 with $n_s \geq n_{\min}$ for all s , let $t \in (0, 1)$, $\pi_t = \mathbb{P}(D_S > t)$ and*
771 *$h_t = 1 - (1 - t)^{n_{\min}}$. Then*

$$\mathbb{P}(\pi_t > U(J, m)/h_t) \leq \delta.$$

772 *Hence, with $\hat{\gamma} = U(J, m)/h_t$, a new camera satisfies $p_{\text{new}} \leq q_{\text{new}} + t$ except*
773 *on a set of cameras of population mass at most $\hat{\gamma}$, with confidence $1 - \delta$.*

774 *Proof.* $\mathbb{P}(B_s = 1) \geq \mathbb{P}(D_S > t)(1 - (1 - t)^{n_s}) \geq \pi_t h_t$. The coupling argument
775 of Proposition A.11 gives $J \succeq_{\text{st}} \text{Bin}(m, \pi_t h_t)$, so $\mathbb{P}(U(J, m) < \pi_t h_t) \leq \delta$,
776 which is the claim. $\square$

777 *Remark A.14* (data requirement; checks). With $J = 0$, $\hat{\gamma} \leq \gamma$ requires $m \geq$
778 $m_{\min}$ with $1 - \delta^{1/m_{\min}} \leq \gamma$ ($m_{\min} = 59$ for $\gamma = 5\%$, 29 for $\gamma = 10\%$) and $n_{\min} \geq$
779 $\ln(1 - U(0, m)/\gamma)/\ln(1 - t)$ events in every scene; the total $m n_{\min}$ decreases
780 to the floor $\approx \ln(1/\delta)/(\gamma t)$ as m grows. *Check:* two-point populations $D_S \in$
781 $\{0, t'\}$, $t' > t$, $m = 48$, 10^5 runs: maximum failure 0.0442.

782 *Appendix A.8. C5a: a pessimistic-length twin is conservative under periodic*
783 *sampling*

784 **Proposition A.15** (C5a). *Let an event occupy frames $F = \{o, \dots, o + d - 1\}$*
785 *and let the detector outcome on each frame be fixed (a deterministic function*
786 *of the frame). For the truncated event $F_L = \{o, \dots, o + L - 1\}$, $L \leq d$,*
787 *with hit set $H_L = H \cap F_L$, the miss probability under a uniform phase,*
788 *$m(L) = 1 - |H_L \bmod K|/K$, is non-increasing in L . Consequently a twin built*
789 *with a pessimistic (shorter) length L has $q_L \geq q$ and discordance $D_L \leq D$,*
790 *so the C3 bound $q_L + D_L$ remains valid and the miss term is conservative.*

791 *Proof.* $L \leq L'$ implies $H_L \subseteq H_{L'}$, hence $H_L \bmod K \subseteq H_{L'} \bmod K$ and
792 $m(L) \geq m(L')$. A twin catch at phase φ for the truncated twin implies
793 a catch for the full twin, so $\{M = 1, M'_L = 0\} \subseteq \{M = 1, M' = 0\}$ and
794 $D_L \leq D$; Lemma A.5 holds for any pairing. $\square$

795 *Remark A.16* (check). On the twin of the 48 static scenes, shortening every
796 event by $x \in \{10, 25, 50\}\%$ gives non-decreasing q for every event and every
797 $K \in \{1, 4, 8, 16, 48\}$ (all monotone: yes); e.g. at $d_{\min} = 32$, $K = 16$, q rises
798 from 1.37% to 2.36% at $x = 50\%$.

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